



COMP30026

Models of Computation

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Lecture 8

Predicate Logic: Clausal Form



Rules for Quantifiers

We cannot always move subformulas through quantifiers.

Consider $\exists x(P(x) \wedge Q(x))$.



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However, for all formulas F and G :

$$\neg \forall x F \equiv \exists x (\neg F)$$

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$$\exists x (F \vee G) \equiv (\exists x F) \vee (\exists x G)$$

$$\forall x (F \wedge G) \equiv (\forall x F) \wedge (\forall x G)$$

From this, you can derive

$$\exists x (F \rightarrow G) \equiv (\forall x F) \rightarrow (\exists x G)$$



More Rules for Quantifiers

If G is a formula with **no free occurrences** of x , then:

$$\exists x G \equiv G$$

$$\forall x G \equiv G$$

$$\exists x(F \wedge G) \equiv (\exists x F) \wedge G$$

$$\forall x(F \vee G) \equiv (\forall x F) \vee G$$

$$\forall x(F \rightarrow G) \equiv (\exists x F) \rightarrow G$$

$$\forall x(G \rightarrow F) \equiv G \rightarrow (\forall x F)$$

no matter what F is (may contain free x).



Resolution for Predicate Logic

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Resolution for Predicate Logic

- Resolution generalises to predicate logic.
- Try to derive $\perp$.
- CNF in predicate logic: conjunction of disjunctions of literals with all variables universally quantified
- **However**, quantifiers mean that not every predicate logic formula has a CNF.
- Work around by giving up on equivalence.



Resolution for Predicate Logic: Idea

Extend Resolving

- From simple literals
- To resolving on atomic formulas
(containing variables, constants, and function symbols)

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Example

“Every lorikeet is gorgeous” and “Coco is not gorgeous” entails “Coco is not a lorikeet.”

Formally:

$$\neg L(x) \vee G(x) \quad \text{and} \quad \neg G(c)$$

gives $\neg L(c)$.

Resolution for Predicate Logic: Idea

Example

“The program a can output everything” and “No program can output b”

$$\forall x.O(a, x) \quad \text{and} \quad \forall y.\neg O(y, b)$$

Or in clausal form

$$O(a, x) \quad \text{and} \quad \neg O(y, b)$$



Resolution by Example

All variables are universally quantified
(globally or **equivalently** per clause):

$$(\neg P(f(a)) \vee R(b)) \wedge (P(f(z)) \vee Q(z, u)) \wedge \neg R(b) \wedge \neg Q(w, f(v))$$

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Then

1. $\neg P(f(a)), R(b)$
2. $P(f(z)), Q(z, u)$
3. $\neg R(b)$
4. $\neg Q(w, f(v))$
5. $Q(a, u), R(b)$ resolve 1, 2 $\{z \mapsto a\}$
6. $Q(a, u)$ resolve 5, 3
7. $\perp$ resolve 4, 6 $\{w \mapsto a, u \mapsto f(v)\}$



Eliminating Existential Quantifiers

Consider $F: \exists x \forall y P(x, y)$.

Consider $G: \forall y P(a, y)$, where a is a constant symbol.

Theorem

G is *satisfiable* iff F is.

Note: For an existential quantifier in front of a formula!



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Proof (sketch)

Every model of G is a model of F : pick $I(a)$ for $\exists x$.

We can extend every model of F to a model of G by defining $I(a)$ to be the same object we picked for $\exists x$.



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- could be a different x for another y ...
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Only Skolemizing top-level existentials is examinable.



Predicate Logic Formula to Clausal Form

The process is similar to what we did with propositional formulas:

1. Eliminate $\leftrightarrow$ and $\rightarrow$.
2. Push negation in. (Bring to NNF.)
3. Rename shadowed variables.
 - 3.1 A variable is **shadowed** iff it is within the scope of a quantifier binding the same variable (e.g. the second x in $\forall x \exists x$ or $\forall x \forall x$).
4. Eliminate existential quantifiers (Skolemize).
5. Drop universal quantifiers (they are still there)
6. Bring to CNF (using the distributive laws).

Clausal Form: Step 1—Use Just $\forall$, $\wedge$, $\neg$

Let us use this running example:

$$\forall x (P(x) \leftrightarrow \exists y (R(x, y) \wedge \forall z R(z, y)))$$

First use the usual translations to eliminate $\leftrightarrow$ and $\rightarrow$:

$$\forall x \left(\begin{array}{l} (P(x) \rightarrow \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge \\ (\exists y (R(x, y) \wedge \forall z R(z, y)) \rightarrow P(x)) \end{array} \right)$$

which then becomes:

$$\forall x \left(\begin{array}{l} (\neg P(x) \vee \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge \\ (\neg \exists y (R(x, y) \wedge \forall z R(z, y)) \vee P(x)) \end{array} \right)$$

Clausal Form: Step 2—Push Negation

Next drive negation in.

$$\forall x \left((\neg P(x) \vee \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge (\neg \exists y (R(x, y) \wedge \forall z R(z, y)) \vee P(x)) \right)$$

then becomes

$$\forall x \left((\neg P(x) \vee \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge (\forall y (\neg R(x, y) \vee \exists z \neg R(z, y)) \vee P(x)) \right)$$

Clausal Form: Step 3—Rename Apart

Now rename variables so that no two quantifiers use the same variable name. With that,

$$\forall x \left(\begin{array}{l} (\neg P(x) \vee \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge \\ (\forall y (\neg R(x, y) \vee \exists z \neg R(z, y)) \vee P(x)) \end{array} \right)$$

turns into, say

$$\forall x \left(\begin{array}{l} (\neg P(x) \vee \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge \\ (\forall u (\neg R(x, u) \vee \exists v \neg R(v, u)) \vee P(x)) \end{array} \right)$$



Clausal Form: Step 4—Skolemize

Let us highlight the existentially quantified variables:

$$\forall x \left((\neg P(x) \vee \exists y (R(x, y) \wedge \forall z R(z, y))) \wedge (\forall u (\neg R(x, u) \vee \exists v \neg R(v, u)) \vee P(x)) \right)$$

The existentially quantified y is in the scope of $\forall x$ —so replace it by $f(x)$.

Clausal Form: Step 4—Skolemize

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The existentially quantified y is in the scope of $\forall x$ —so replace it by $f(x)$.

The existentially quantified v is in the scope of $\forall x$, as well as of $\forall u$. So we replace it by $g(u, x)$.

$$\forall x \left(\begin{array}{l} (\neg P(x) \vee (R(x, f(x)) \wedge \forall z R(z, f(x)))) \wedge \\ (\forall u (\neg R(x, u) \vee \neg R(g(u, x), u)) \vee P(x)) \end{array} \right)$$

Clausal Form: Step 5—Drop Universal Quantifiers

Eliminating universal quantifiers is easy:

$$\forall x \left(\begin{array}{l} (\neg P(x) \vee (R(x, f(x)) \wedge \forall z R(z, f(x)))) \wedge \\ (\forall u (\neg R(x, u) \vee \neg R(g(u, x), u)) \vee P(x)) \end{array} \right)$$

becomes

$$(\neg P(x) \vee (R(x, f(x)) \wedge R(z, f(x)))) \wedge (\neg R(x, u) \vee \neg R(g(u, x), u) \vee P(x))$$

(The quantifiers are still implicitly there)

Clausal Form: Step 6—Convert to CNF

$$(\neg P(x) \vee (R(x, f(x)) \wedge R(z, f(x)))) \wedge (\neg R(x, u) \vee \neg R(g(u, x), u) \vee P(x))$$

Distribute $\wedge$ and $\vee$:

$$\begin{aligned} &(\neg P(x) \vee R(x, f(x))) \wedge \\ &(\neg P(x) \vee R(z, f(x))) \wedge \\ &(\neg R(x, u) \vee \neg R(g(u, x), u) \vee P(x)) \end{aligned}$$

Equivalently as a set of literals:

$$\left\{ \begin{array}{l} \{\neg P(x), R(x, f(x))\}, \\ \{\neg P(x), R(z, f(x))\}, \\ \{\neg R(x, u), \neg R(g(u, x), u), P(x)\} \end{array} \right\}$$



Overview

Today

- Idea behind predicate logic resolution
- CNF for predicate logic

Next

- Resolution formally
- Unification
- More examples